**AGA KHAN UNIVERSITY EXAMINATION BOARD**

**SECONDARY SCHOOL CERTIFICATE**

**CLASS X EXAMINATION**

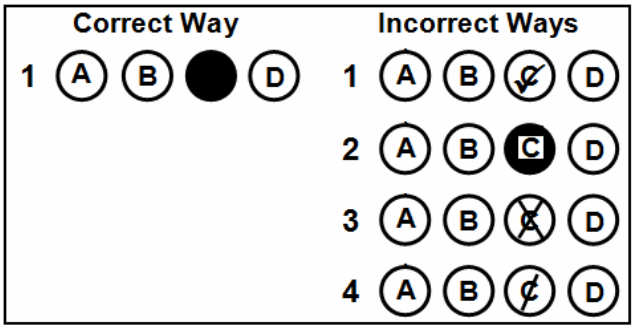
**MAY 2013**

**Mathematics Paper I**

**Time: 40 minutes Marks: 30**

**INSTRUCTIONS**

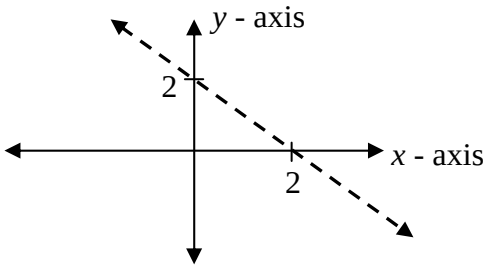
1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.




**Candidate's Signature**

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. The given graph represents the line




A. $y = mx + 2$
B. $y = mx$
C. $y = 2$
D. $x = 2$

2. For the equation $y = 2x - 3$ the abscissa is zero. So, the ordinate will be

A. $-3$
B. $-2$
C. $2$
D. $3$

3. On simplifying, $\frac{ab+b^2}{(a+b)^2}$ is equal to

A. $b$
B. $\frac{b}{a}$
C. $\frac{b}{(a+b)}$
D. $\frac{a}{(a+b)}$

4. For $x = 3$ and $x + y = 1$, the solution set is

A. $\{(-3, 2)\}$
B. $\{(3, -2)\}$
C. $\{3, -2\}$
D. $\{-3, 2\}$

5. The highest common factor (H.C.F) of $(x+y)^2$ and $x^2 + xy$ is

A. $(x+y)^2$
B. $x^2 + xy$
C. $(x+y)$
D. $x(x+y)^2$

6. The least common multiple (L.C.M) of 1 and $\left(x^2 - y^2\right)$ is

A. 1
B. $(x - y)$
C. $(x + y)$
D. $\left(x^2 - y^2\right)$

7. The standard form of the quadratic equation $(x - a)(x - b) = 0$ is

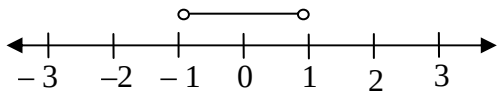
A. $x^2 + (a - b)x + ab = 0$
B. $x^2 - (a - b)x - ab = 0$
C. $x^2 - (a + b)x - ab = 0$
D. $x^2 - (a + b)x + ab = 0$

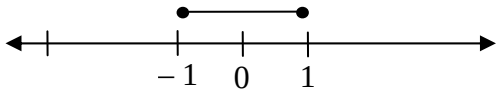
8. If $x + 2 = A(x + 1) + B(x)$, then the value of $A$ will be

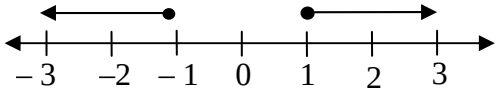
A. $-2$
B. $-1$
C. 1
D. 2

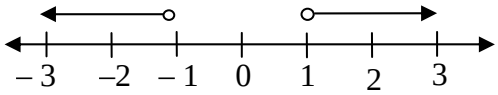
9. The solution set of $|x| < 1$ may be represented on the number line as follows

**Note:** $x \in R$

A.



B.



C.



D.



10. The solution set of $x + 2 \leq 2$ is

**Note:** $x \in \omega$

A. $\{\ \}$
B. $\{0\}$
C. $\{0, 1, 2, 3, 4.....\}$
D. $\{.....-4, -3, -2, -1, 0, .....\}$

11. For the equation $x^2 - 2x + 2 = 0$, the value of the discriminant $(b^2 - 4ac)$ is equal to

A. $-12$
B. $-4$
C. $4$
D. $12$

12. If the solution set of $\sqrt{x} = -a$ exists, then which of the following statements is correct?

I. $a > 0$
II. $a < 0$
III. $a = 0$

A. I only
B. II only
C. I and III only
D. II and III only

13. If $a > 5$, then which of the following options is correct?

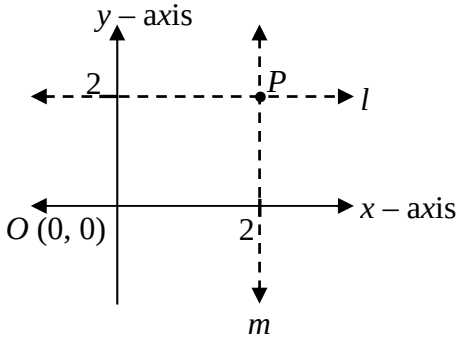
A. $\frac{1}{a} < -\frac{1}{5}$
B. $\frac{1}{a} \geq -\frac{1}{5}$
C. $\frac{1}{a} < \frac{1}{5}$
D. $\frac{1}{a} > \frac{1}{5}$

14. If $ax^2 - b = 0$, then its solution set will be

A. $\left\{\pm\frac{b}{a}\right\}$
B. $\left\{\pm\frac{a}{b}\right\}$
C. $\left\{\pm\sqrt{\frac{a}{b}}\right\}$
D. $\left\{\pm\sqrt{\frac{b}{a}}\right\}$

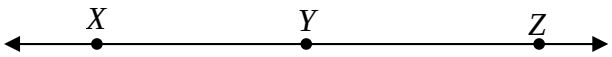
15. In the given figure, if $P$ is the point of intersection of the line $l$ and $m$, then $m\overline{OP}$ is equal to

A. 2 units.
B. 4 units.
C. 8 units.
D. $2\sqrt{2}$ units.



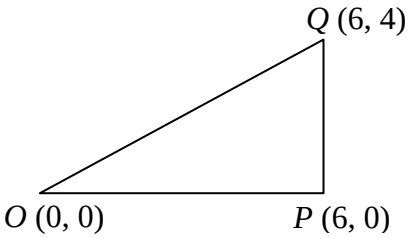

16. If $X$, $Y$ and $Z$ are collinear points as shown in the figure, then which of the following conditions is correct?

A. $|XZ| = |XY| + |YZ|$
B. $|XY| = |XZ| + |YZ|$
C. $|XY| = |YZ| - |XY|$
D. $|XZ| = |XY| - |YZ|$




17. In the given triangle $OPQ$, the mid-point of the side opposite to vertex $P$ is

A. (3, 0)
B. (6, 2)
C. (3, 2)
D. (3, 3)




**NOT TO SCALE**

18. The angle of measure $16°72'$ can be written as

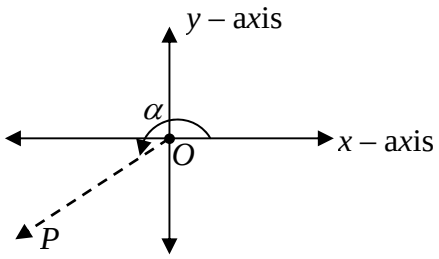
A. $17°12'$
B. $(16.72)°$
C. $(17.12)°$
D. $16°.72'$

19. $\sin^2\theta \operatorname{cosec}\theta$ is equal to

A. $\cos\theta$
B. $\sin\theta$
C. $\sin\theta\tan\theta$
D. $\cos\theta\cos\theta$

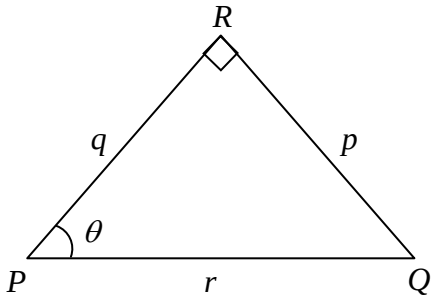
20. For the terminal ray $\overrightarrow{OP}$ as shown in the diagram, which of the following conditions is true?

A. $\sin\alpha > 0$ and $\cos\alpha < 0$
B. $\sin\alpha < 0$ and $\cos\alpha > 0$
C. $\cos\alpha < 0$ and $\tan\alpha > 0$
D. $\cos\alpha > 0$ and $\tan\alpha < 0$




21. In the right-angled triangle $PQR$, $\sin^2\theta$ is equal to

A. $\frac{p}{r}$
B. $\frac{p^2}{r^2}$
C. $\frac{p}{q}$
D. $\frac{p^2}{q^2}$

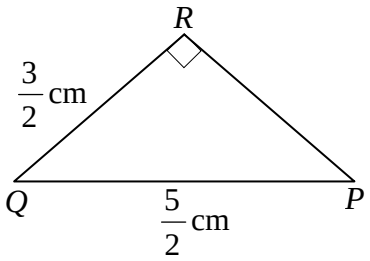



22. In a right-angled triangle if the length of the shortest and the longest sides are 6 cm and 10 cm respectively, then the length of the third side will be

A. 6 cm.
B. 7 cm.
C. 8 cm.
D. 9 cm.

23. In the given right-angled triangle $m\overline{PR}$ is equal to

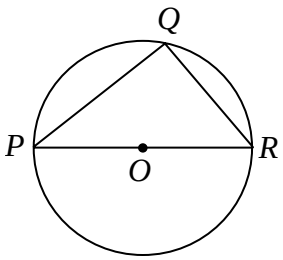
A. $\frac{3}{2}$cm.
B. $\frac{7}{2}$ cm.
C. 2 cm.
D. 4 cm.




**NOT TO SCALE**

24. In the circle with centre $O$ as shown in the diagram if $m\overline{PQ} = m\overline{QR} = 2\sqrt{2}$, then the length of $\overline{PR}$ is

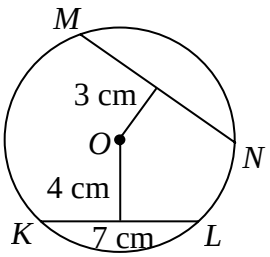
A. $4\sqrt{2}$ units.
B. 16 units.
C. 8 units.
D. 4 units.




**NOT TO SCALE**

25. Consider two chords $\overline{KL}$ and $\overline{MN}$ in a circle with centre $O$. If $KL$ and $MN$ are at a distance of 4 cm and 3 cm respectively, then

A. $m\,\overline{MN} > 7$ cm.
B. $m\,\overline{MN} < 7$ cm.
C. $m\,\overline{MN} \geq 7$ cm.
D. $m\,\overline{MN} \leq 7$ cm.

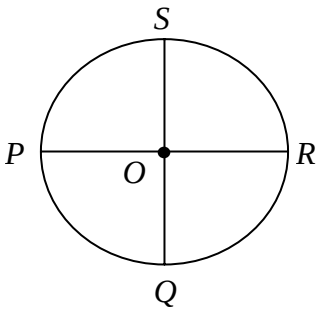



**NOT TO SCALE**

26. Consider the given circle with centre $O$. Which of the following statements is/ are correct?

I. $m\,\overline{OP} = m\,\overline{OR} \neq m\,\overline{OQ}$
II. $m\,\overline{OP} = m\,\overline{OQ}$
III. $m\overline{OP} =$ radius of the circle

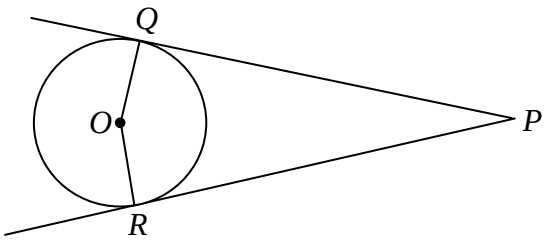
A. I only
B. II only
C. II and III only
D. I and III only




27. If in the given figure $PQ$ and $PR$ are tangents to the circle, then which of the following options is correct?

I. $\overline{OQ}$ perpendicular to $\overrightarrow{PQ}$
II. $\overline{PQ} \cong \overline{PR}$
III. $\overline{PO} \cong \overline{PQ} \cong \overline{PR}$

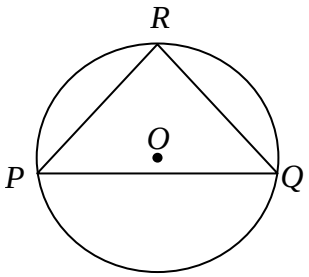
A. I only
B. II only
C. I and III only
D. I and II only



28. If $PQ$ is a chord and not passing through the centre $O$ of the given circle, then $\angle PRQ$ is

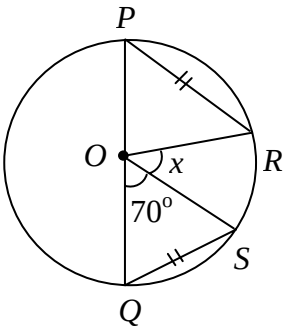
I. greater than $90°$.
II. less than $90°$.
III. equal to $90°$.

A. I only
B. II only
C. I and III only
D. II and III only




29. In the given figure, if $O$ is the centre of circle and $m\angle QOS = 70°$, then $x$ is equal to

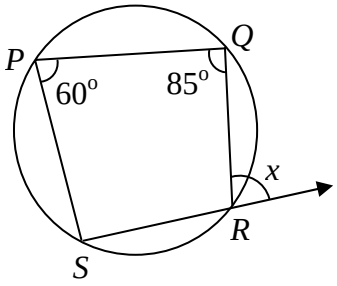
A. 30°
B. 40°
C. 60°
D. 70°




**NOT TO SCALE**

30. In the given figure, the value of $x$ is equal to

A. 60°
B. 85°
C. 95°
D. 120°




**NOT TO SCALE**

END OF PAPER